

ATE Survey — User Guide

This guide walks you through configuring and running the ATE Survey workflow, which processes Amboseli Trust for Elephants community questionnaire responses to produce attitude charts, sentiment scores, statistical analyses, spatial maps, and a Word report.

Overview

The workflow delivers, for each run:

- **2 Likert charts** — community attitudes/beliefs (agree/disagree) and mitigation effectiveness ratings
 - **26 pie charts** and **9 bar charts** across demographic, knowledge, and behaviour questions
 - **Elephant sentiment scores** — a per-respondent overall attitude score computed from 11 belief columns
 - **ANOVA, boxplots, OLS scatter plots, and Tukey HSD plots** — statistical analysis of sentiment vs demographic factors
 - **3 spatial maps** — survey locations, gender distribution, and attitude scores overlaid on Amboseli ranch boundaries
 - **1 Word report** (`social_survey.docx`) populated with all charts and a demographic summary table
-

Prerequisites

Before running the workflow, ensure you have:

- Access to an **EarthRanger** instance with `atequestionnaire_rep` events recorded for the analysis period

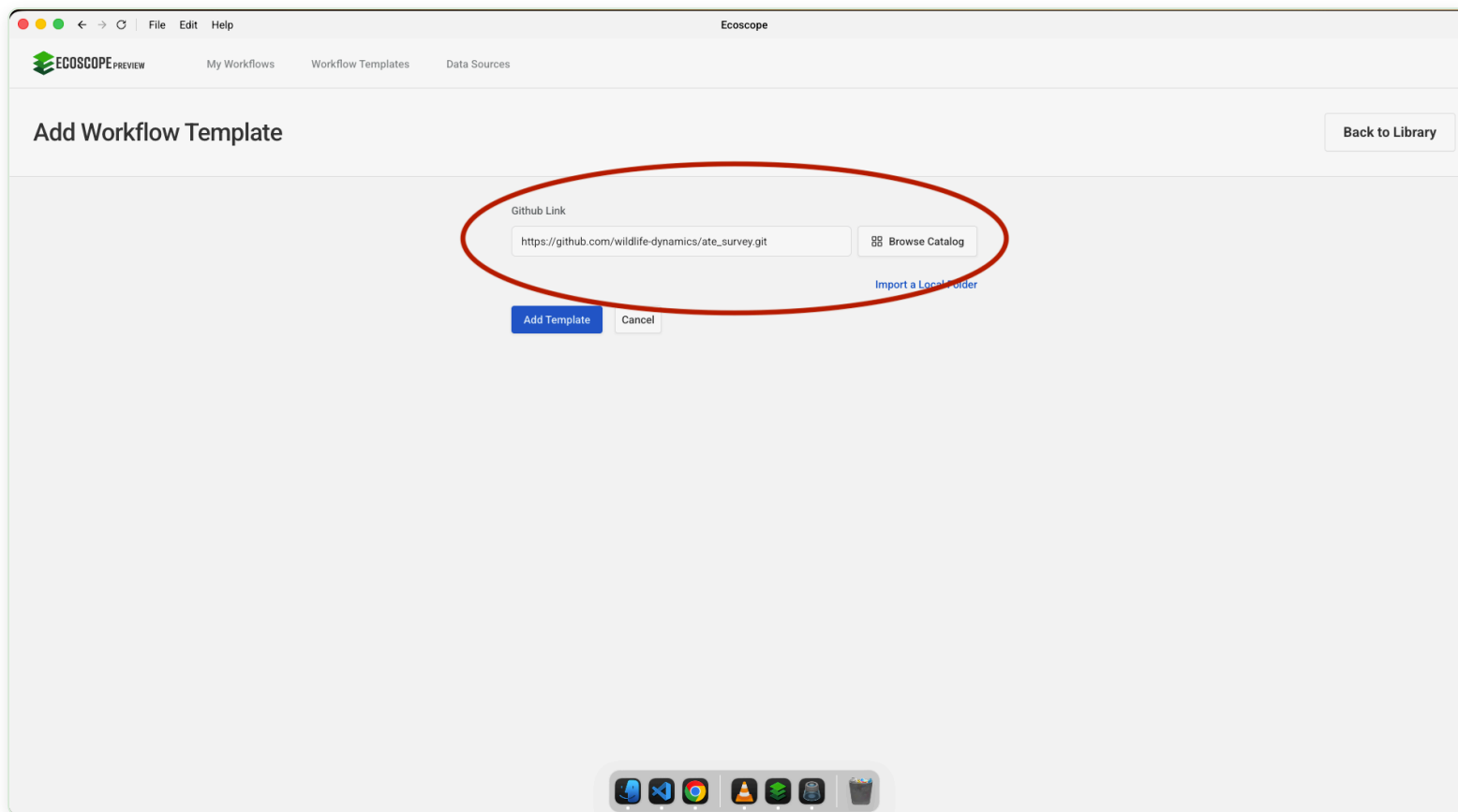
Step-by-Step Configuration

Step 1 — Add the Workflow Template

In the workflow runner, go to **Workflow Templates** and click **Add Workflow Template**. Paste the GitHub repository URL into the **Github Link** field:

```
https://github.com/wildlife-dynamics/ate_survey.git
```

Then click **Add Template**.



Add Workflow Template

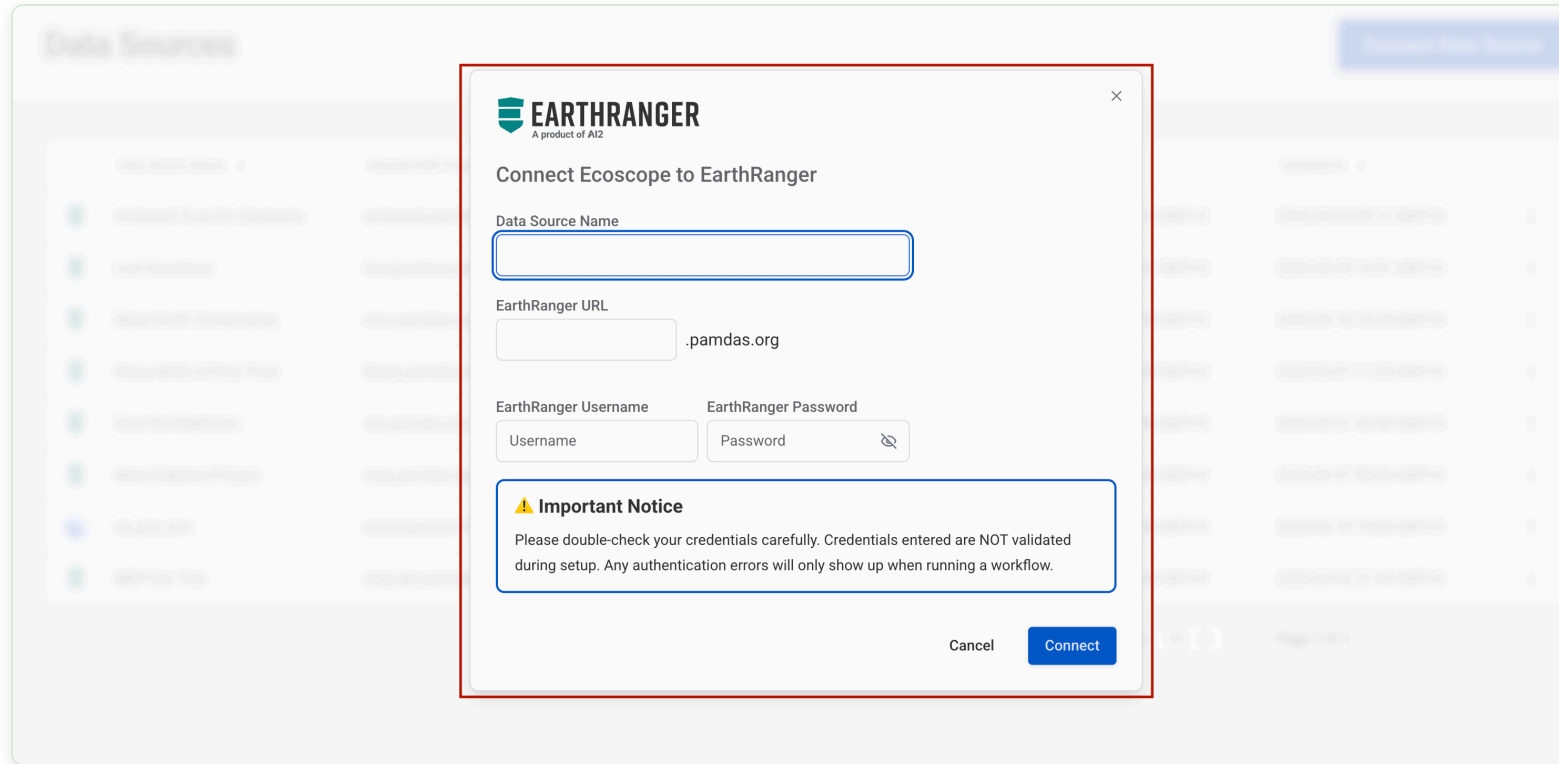
Step 2 — Add an EarthRanger Connection

Navigate to **Data Sources** and click **Connect**, then select **EarthRanger**. Fill in the connection form:

- **Data Source Name** — a label to identify this connection (e.g. Amboseli Trust for Elephants)
- **EarthRanger URL** — your instance URL (e.g. your-site.pamdas.org)
- **EarthRanger Username** and **EarthRanger Password**

Credentials are not validated at setup time. Any authentication errors will appear when the workflow runs.

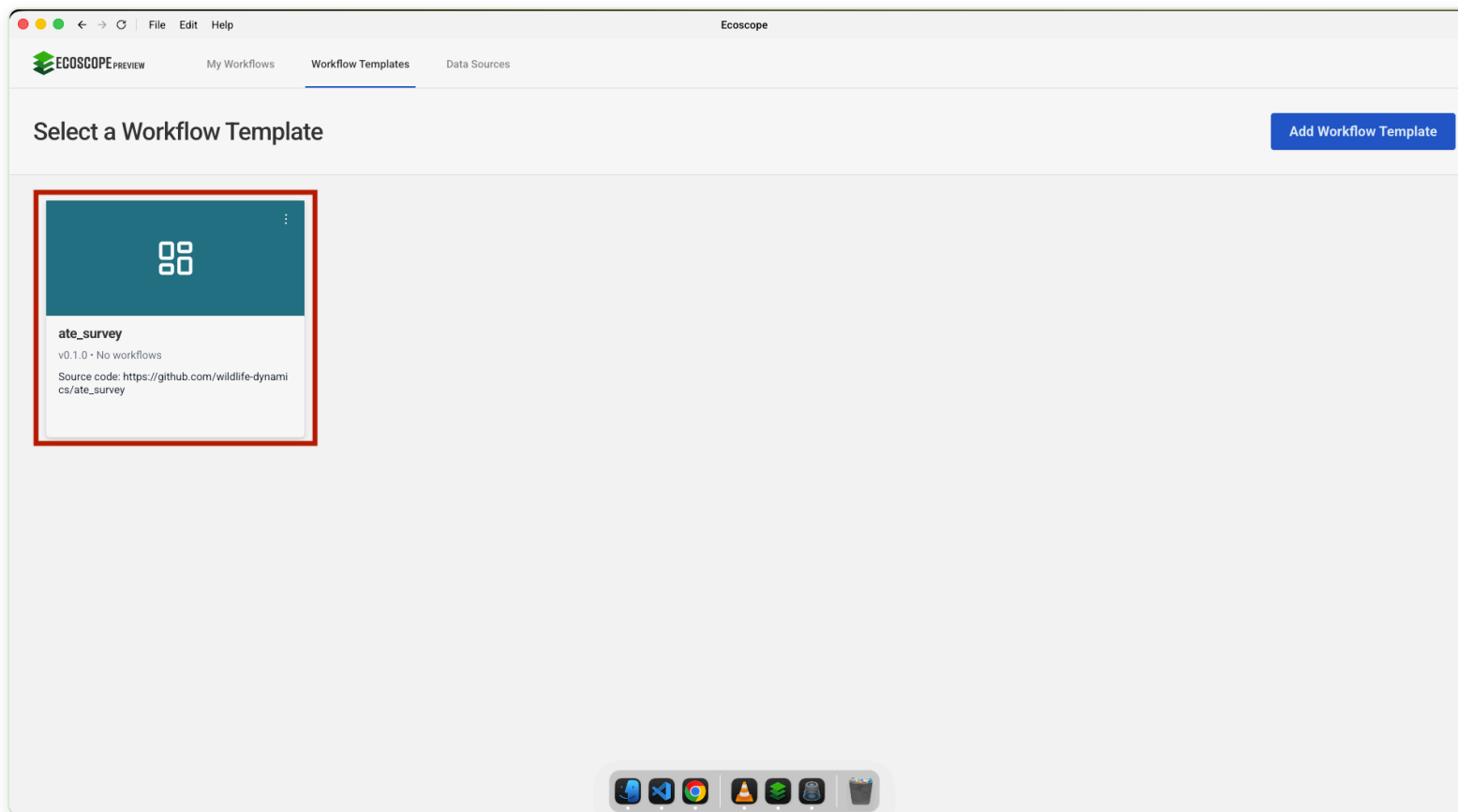
Click **Connect** to save.

The image shows a modal dialog box titled "Connect Ecoscope to EarthRanger" with the EarthRanger logo at the top left. The dialog contains several input fields: "Data Source Name" (a text box), "EarthRanger URL" (a text box with ".pamdas.org" entered), "EarthRanger Username" (a text box with "Username" entered), and "EarthRanger Password" (a password box with "Password" entered and a toggle icon). Below these fields is a blue-bordered box with a warning icon and the text "Important Notice: Please double-check your credentials carefully. Credentials entered are NOT validated during setup. Any authentication errors will only show up when running a workflow." At the bottom right are "Cancel" and "Connect" buttons. The background is a blurred view of the "Data Sources" page.

EarthRanger Connection

Step 3 — Select the Workflow

After the template is added, it appears in the **Workflow Templates** list as **ate_survey**. Click the card to open the workflow configuration form.



Select Workflow Template

Steps 4–6 — Set Workflow Details, Time Range, and Connect to EarthRanger

The configuration form has three sections, all visible on a single page.

Step 4 — Set workflow details

Field	Description
Workflow Name	A short name to identify this run

Field	Description
Workflow Description	Optional notes (e.g. survey period or site)

Step 5 — Set time range

Field	Description
Timezone	Select the local timezone (e.g. <code>Africa/Nairobi UTC+03:00</code>)
Since	Start date and time — all <code>atequestionnaire_rep</code> events from this point are fetched
Until	End date and time of the analysis window

Step 6 — Connect to ER

Select the EarthRanger data source configured in Step 2 from the **Data Source** dropdown (e.g. `Amboseli Trust for Elephants`).

Once all three sections are filled, click **Submit**.

The screenshot shows the Ecoscope web application interface. On the left is a sidebar with three menu items: "Set workflow details", "Set time range", and "Connect to ER". The main content area is divided into three sections, each outlined with a red border:

- Set workflow details:** Includes a text input for "Workflow Name*" and a text area for "Workflow Description".
- Set time range:** Includes a dropdown for "Timezone *" (set to "Africa/Nairobi (UTC+03:00)"), a date-time input for "Since *" (set to "06/03/2026, 12:00 AM"), and a date-time input for "Until *" (set to "05/04/2026, 11:59 PM").
- Connect to ER:** Includes a dropdown for "Data Source*" (set to "Amboseli Trust for Elephants").

At the bottom right are "Cancel" and "Submit" buttons. At the bottom center is a row of application icons.

Set Workflow Details, Time Range, and Connect to EarthRanger

Running the Workflow

Once submitted, the runner will:

1. Fetch `atequestionnaire_rep` events for the analysis period; exclude invalid records; flatten and rename survey columns.
2. Fill missing values; map response strings to display labels (Agree/Disagree, True/False, Effective/Not effective).
3. Bin continuous columns into 5 equal-width bins; produce a demographic summary table.

4. Generate 2 Likert charts, 26 pie charts, and 9 bar charts.
5. Compute per-respondent elephant sentiment scores from 11 attitude columns.
6. Run Type II ANOVA, boxplots, OLS scatter plots, and Tukey HSD post-hoc plots.
7. Download spatial layers from Dropbox (ranch boundaries, swamps, national parks) and render 3 EcoMaps.
8. Download the Word template from Dropbox; populate it with all charts and the demographic table; save as `social_survey.docx`.
9. Save all outputs to the directory specified by `ECOSCOPE_WORKFLOWS_RESULTS`.

Output Files

All outputs are written to `$ECOSCOPE_WORKFLOWS_RESULTS/`. Files marked with `<column>` are produced once per chart column.

File	Description
<code>demographic_table.csv</code>	Summary table: gender, age, tribe, household size, education
<code>elephant_sentiment_scores.csv</code>	Per-respondent overall attitude score
<code>anova_results.csv</code>	Type II ANOVA: sentiment vs gender, age group, education, marital status
<code>elephants_relationship_likert.html / .png</code>	Agree/Disagree Likert chart (11 attitude columns)
<code>effectiveness_mitigation_methods.html / .png</code>	Effective/Not effective Likert chart (3 mitigation columns)
<code><column>_pie.html / .png (x26)</code>	One pie chart per survey question
<code><column>_bar.html / .png (x9)</code>	One bar chart per survey question

File	Description
<column>_boxplot.html / .png (×5)	Sentiment score distribution by demographic group
<column>_ols.html / .png (×2)	OLS regression: age and household size vs sentiment
<column>_tukey.html / .png (×4)	Tukey HSD post-hoc comparison per demographic factor
survey_locations_ecomap.html / .png	All survey point locations on Amboseli base map
gender_distribution_ecomap.html / .png	Survey points coloured by respondent gender
attitude_scores_community.html / .png	Survey points coloured by overall attitude score
social_survey.docx	Final Word report with all charts and demographic table